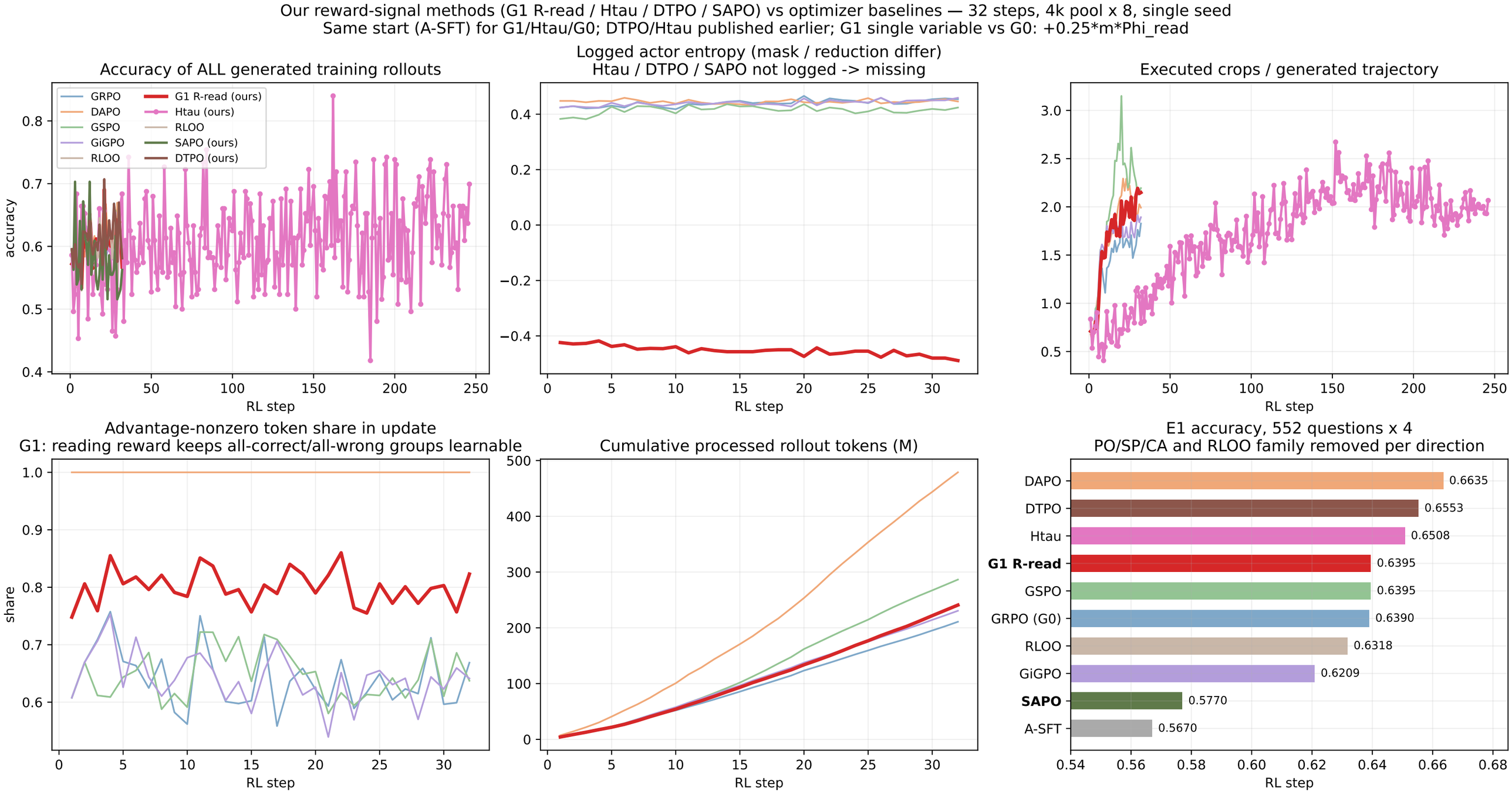
Our reward-signal methods (G1 R-read / Htau / DTPO / SAPO) vs optimizer baselines — 32 steps, 4k pool x 8, single seed
Same start (A-SFT) for G1/Htau/G0; DTPO/Htau published earlier; G1 single variable vs G0: +0.25*m*Phi_read
Accuracy of ALL generated training rollouts
GRPO
DAPO
GSPO
GiGPO
RLOO
G1 R-read (ours)
Htau (ours)
RLOO
SAPO (ours)
DTPO (ours)
accuracy
RL step
Logged actor entropy (mask / reduction differ)
Htau / DTPO / SAPO not logged -> missing
RL step
Executed crops / generated trajectory
RL step
Advantage-nonzero token share in update
G1: reading reward keeps all-correct/all-wrong groups learnable
share
RL step
Cumulative processed rollout tokens (M)
RL step
E1 accuracy, 552 questions x 4
PO/SP/CA and RLOO family removed per direction
DAPO 0.6635
DTPO 0.6553
Htau 0.6508
G1 R-read 0.6395
GSPO 0.6395
GRPO (G0) 0.6390
RLOO 0.6318
GiGPO 0.6209
SAPO 0.5770
A-SFT 0.5670
RL step